

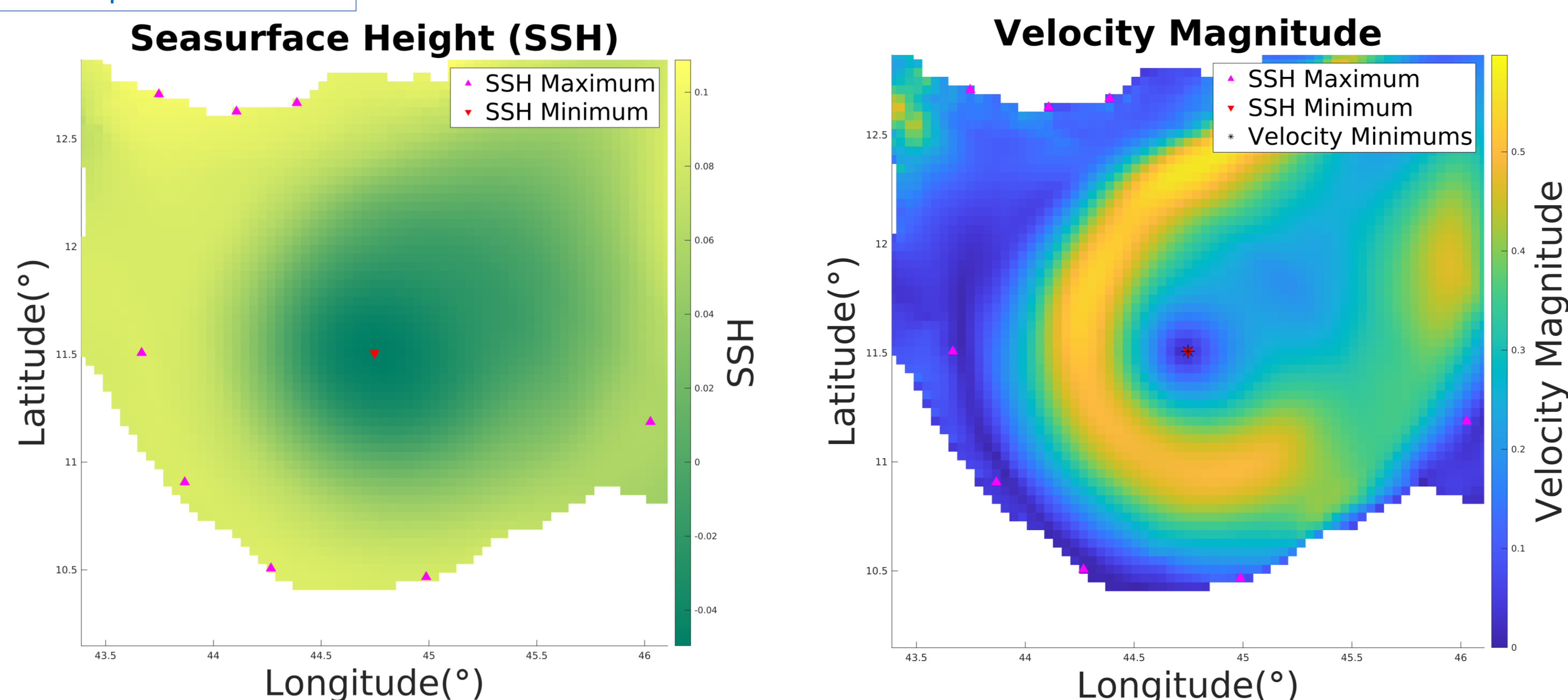
Why We Need To detect Eddies?

- ◇ An eddy is a coherently spinning structure playing an important role in understanding the ocean dynamics
- ◇ Eddies transport salinity, temperature and nutrients vertically and horizontally
- ◇ Most detection methods are designed for 2D satellite data, leaving 3D datasets still not fully discovered

Method Overview

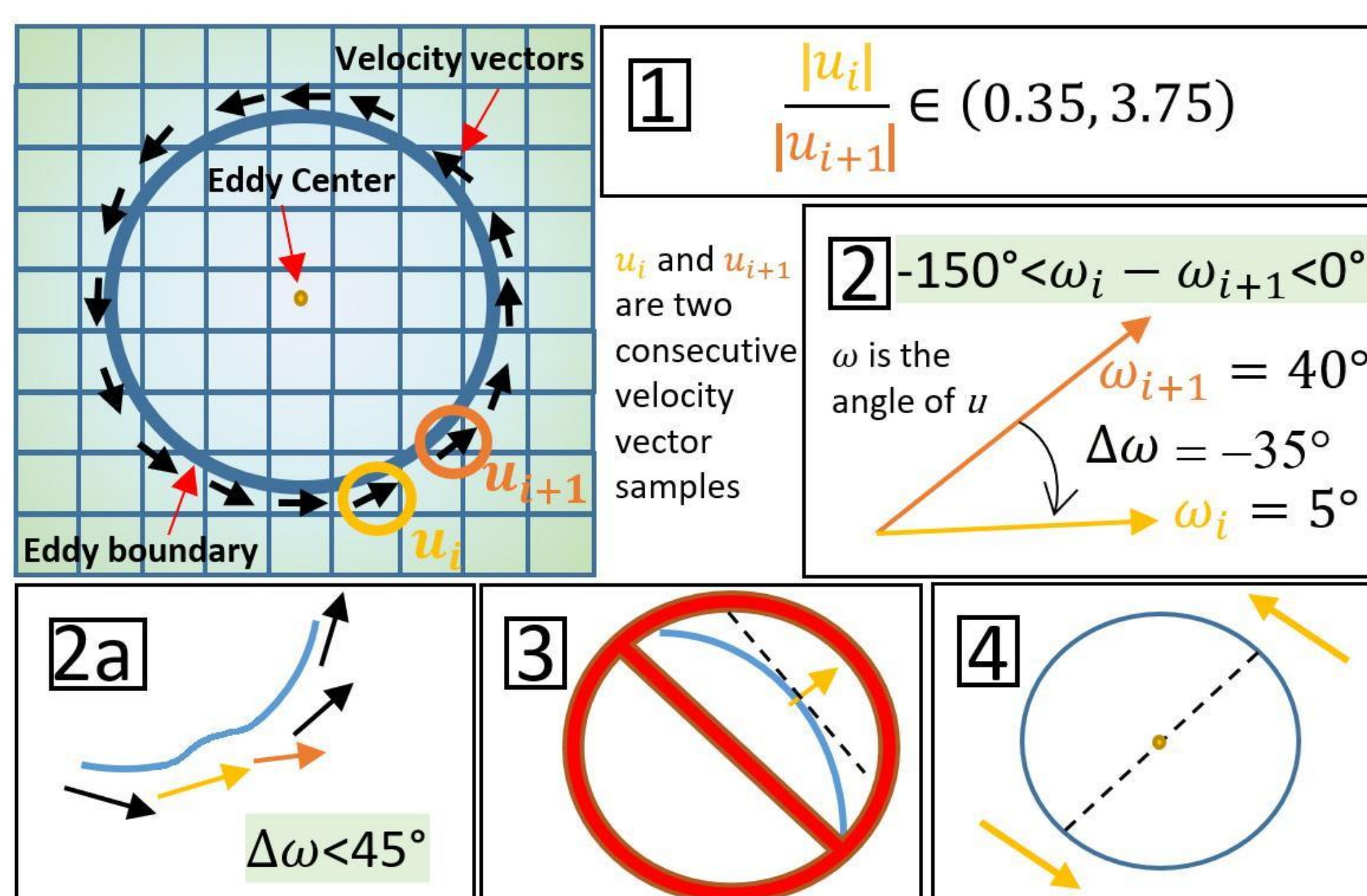
- (A) **Locate potential centers**: Locate the candidate centers of eddies on the sea surface using SSH extrema and velocity magnitude minima.
- (B) **Verify the centers as eddies**: Verify each candidate center with certain conditions to remove streams and other non-rotating or non-symmetrical features from the center candidates.
- (C) **Extract whole structures**: Extract the boundary structure of each eddy starting from each center of eddy (in a region growing fashion).

(A) Locate potential centers



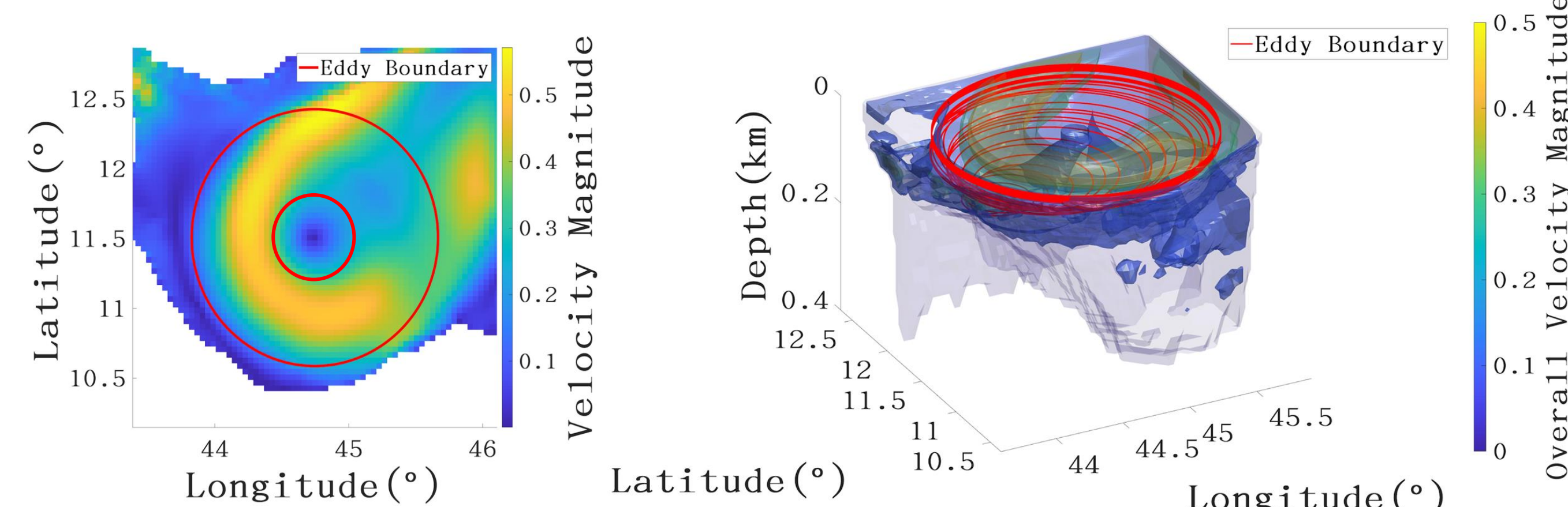
- ◇ Find a local extremum of SSH (sea surface height) first.
- ◇ Set a search region around each SSH extremum.
- ◇ Locate the minimum of the velocity magnitude around it.
- ◇ The first figure on the left is a sample of searching SSH extremum; the second figure is a sample of searching for the velocity minimum.

(B) Verify the centers as eddies



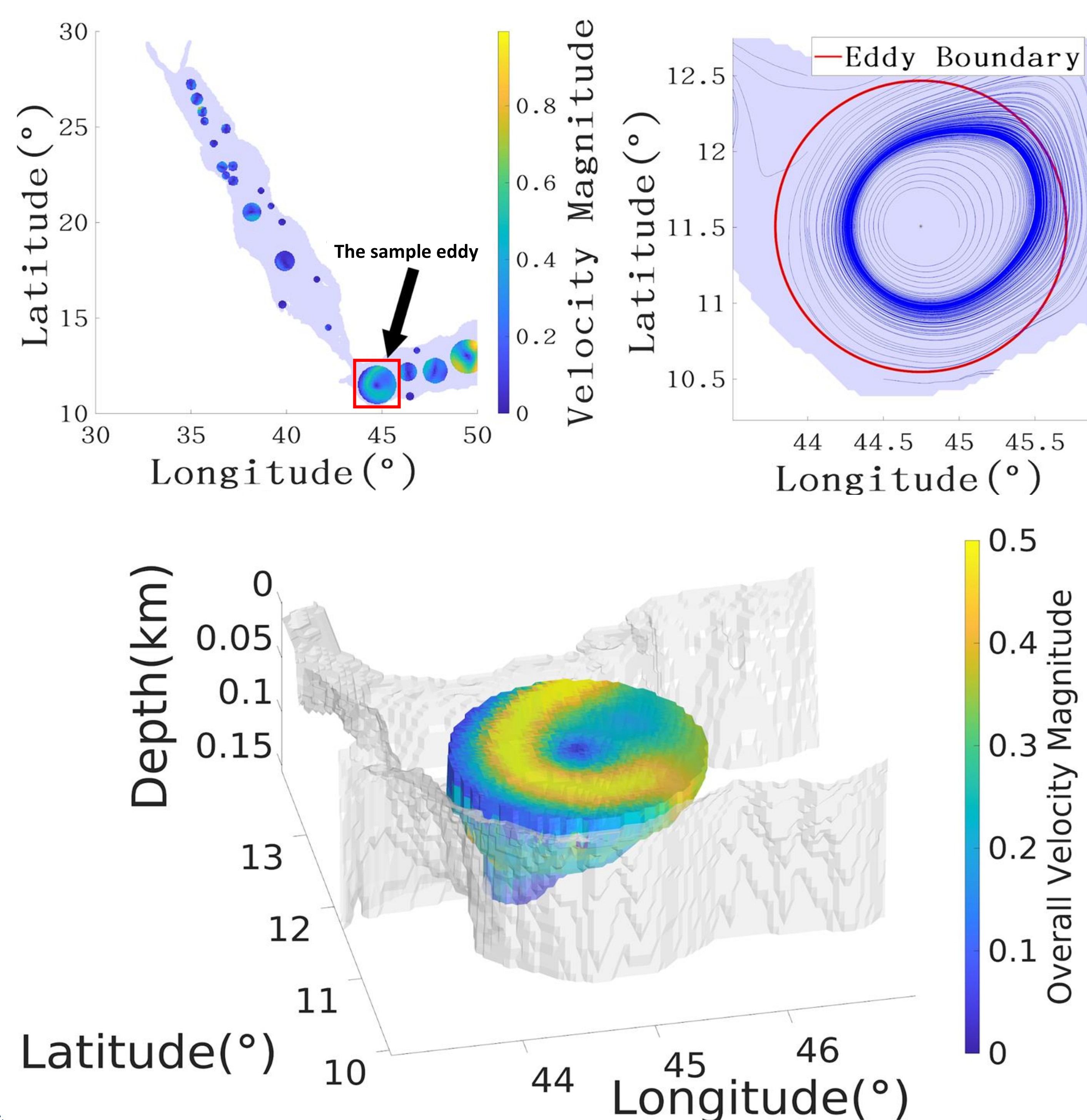
- ◇ Multiple conditions are used to verify potential centers as being actual eddy centers.
- ◇ Those conditions include limits on values, ranges and changes in net velocity, angle and symmetry.

(C) Extract whole structures



- ◇ The rest of the eddy structure will be extracted on each layer by repeating the verification step (b) with larger and larger radius.
- ◇ The center of the eddy in the next layer is searched starting from the neighborhood of the center in current layer along the z axis.

Results

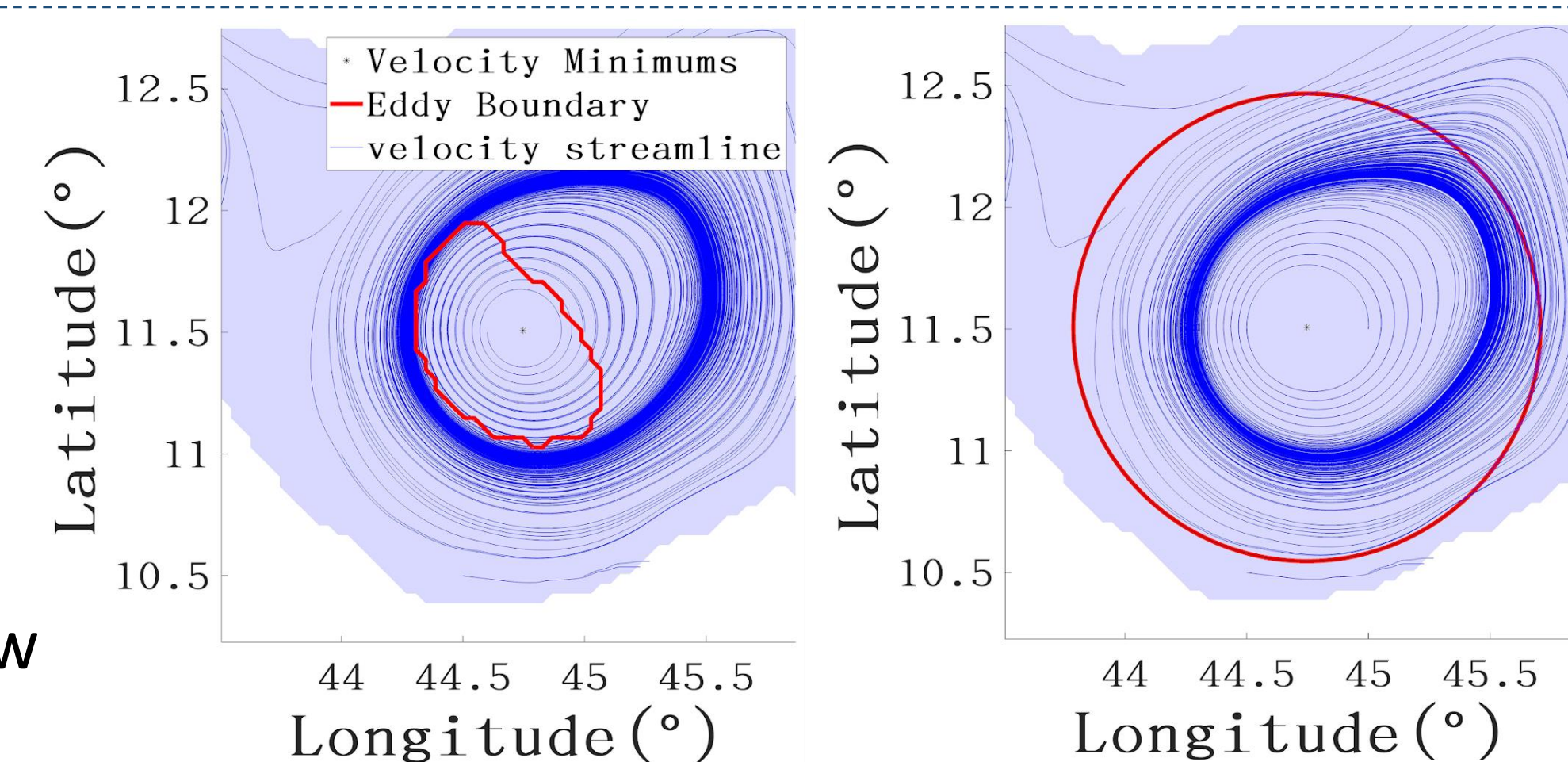


Data Set

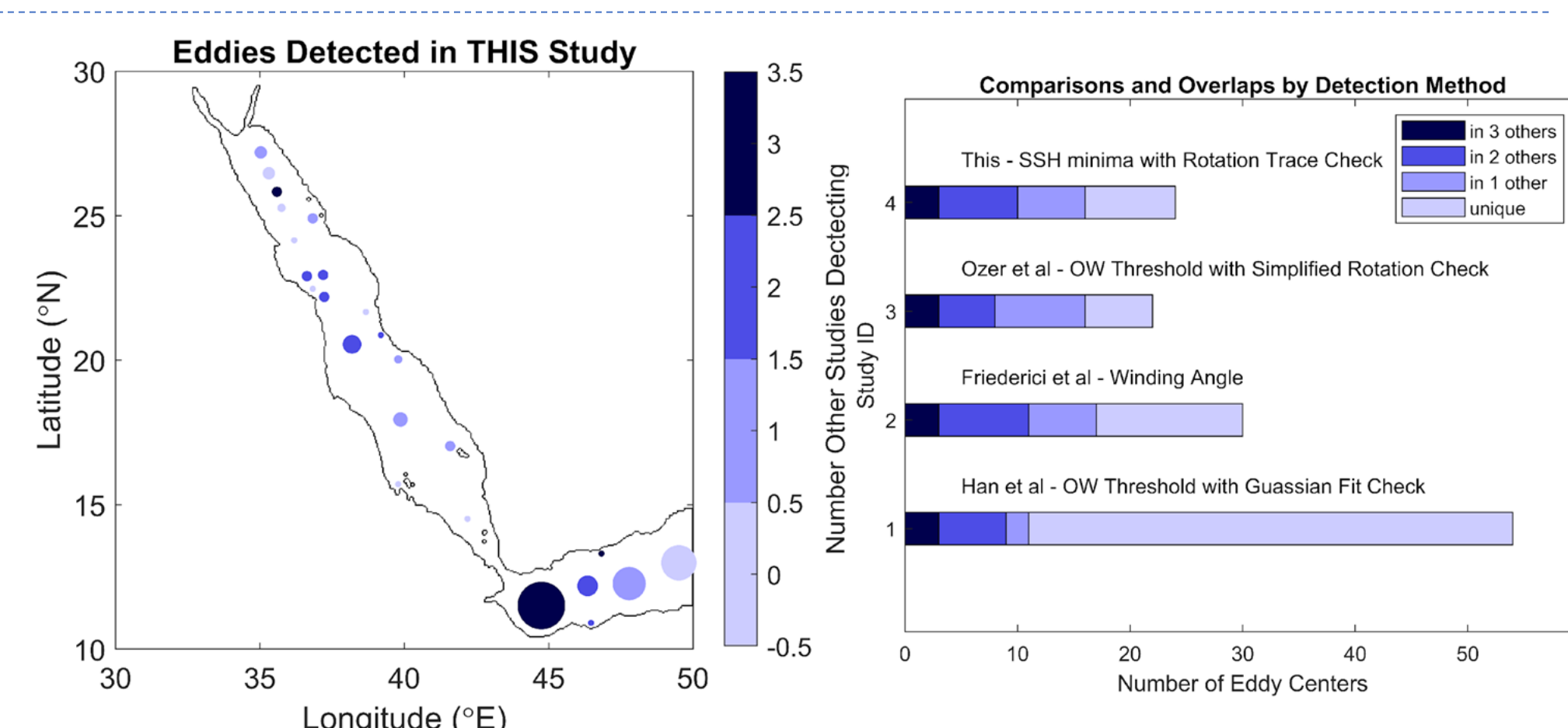
The dataset is a Red Sea simulation data from IEEE VIS Scivis Contest 2020 including 60 ensembles from a one-month period.*

Evaluation

The result on the left is from our previous method using OW parameter while the figure on the right is the output of our new method.



We also compare our method with other methods in Scivis contest 2020 and see a relatively good overlap in the detection results.



Conclusion

We propose a new hybrid eddy extraction approach combining the location of SSH and velocity minima with using the velocity field to assess rotation coherence. Comparison with previous approaches on the Red Sea dataset confirmed both importance of the rotation checks and the sensitivity of results to the method used, value thresholds and criteria.